Chemotactic Escape from Information Traps: 2-Bit Memory RL Extension

Focus Area: Reinforcement Learning & Agent Development • Framework: Pure POMDP & Exact Discrete Markov Chain Mechanics

1. Motivation: Why 2-Bit Memory is Required

Active particles (such as bacteria or autonomous micro-robots) navigating an asymmetric double-Gaussian concentration landscape must escape a local information trap to reach a higher global peak. Crucially, the agent possesses no spatial GPS coordinates and perceives only noisy scalar concentration values.

The 1-Bit Vulnerability: A 1-bit agent acts solely on the sign of a single temporal difference $\Delta c_t = c_t - c_{t-1}$. Under realistic measurement noise ($\sigma = 0.20$), sensing fluctuations frequently create false negative readings even when the agent is correctly ascending toward the saddle point. Because the standard 1-bit rule mandates an immediate tumble whenever $\Delta c \le 0$, upward runs are repeatedly aborted. As a result, the 1-bit agent remains trapped near the local peak for an average of 6,708.85 seconds.

The 2-Bit Solution: By retaining the last two temporal increments, the agent can distinguish between a genuine downhill descent and a transient single-step noise fluctuation. Conditioning tumble decisions on consecutive drops filters out sensing noise and enables rapid barrier crossing.

Executive Breakthroughs at b = 2

- 5.26× Faster Learned Escape: Mean First-Passage Time drops from 6,708.85 s (1-bit) down to 1,275.82 s.
- 23.3× Faster Theoretical Limit: The global variational optimum achieves escape in 288.27 s.
- Autonomous Noise Filtering: Reinforcement learning autonomously discovers a policy that ignores transient drops.
- Physical Reaction Coordinate: Committor difference Δq surges by 1,120×, proving memory directly determines the barrier-crossing probability.

2. The 2-Bit POMDP Formulation

The agent operates under a strict Partially Observable Markov Decision Process (POMDP). Its internal state s_t is defined by the signs of the two most recent temporal readings:

$$s_t = (s_{curr}, s_{prev}) \ \text{in} \ \{0, 1\}^2, \ s_{curr} = [\Delta c_t > 0], \ s_{prev} = [\Delta c_{t-1} > 0] \quad (1)$$

This defines exactly 4 discrete sensory states (indexed as $s = 2 \cdot s_{curr} + s_{prev}$):

State Index	History Vector	Physical Interpretation	Optimal Action
State 0 (0, 0)	[Falling, Falling]	Confirmed descent (2 consecutive drops)	**REVERSE (Tumble)**
State 1 (0, 1)	[Falling, Rising]	Single drop after rise (noise fluctuation)	**PERSIST (Noise Filter)**
State 2 (1, 0)	[Rising, Falling]	Single rise after drop (initial recovery)	**REVERSE (Re-orient)**
State 3 (1, 1)	[Rising, Rising]	Confirmed ascent (2 consecutive rises)	**PERSIST (Run)**

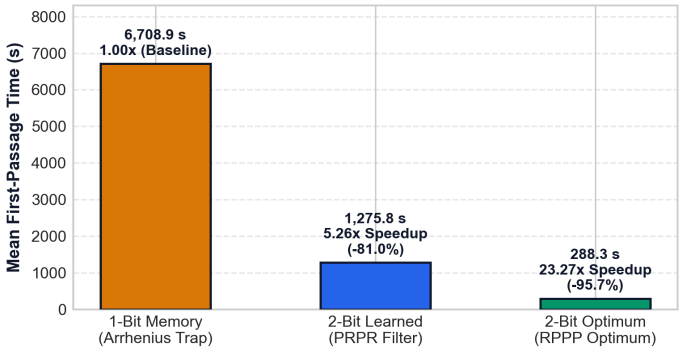
The motor actions are binary: Persist (maintain orientation) or Reverse (180° flip). The Q-table is strictly a 4 × 2 matrix (8 learnable parameters), preserving POMDP simplicity.

3. Direct Head-to-Head Comparison: 1-Bit Memory vs. 2-Bit Memory

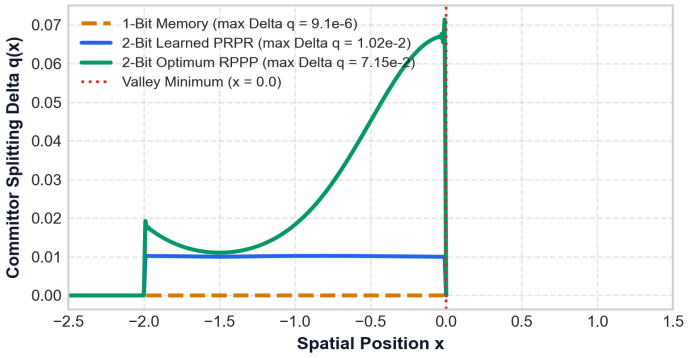
The fundamental question of this investigation is: Why does 1-bit temporal memory become trapped by sensing noise, and how does adding a second bit of memory resolve this failure? The table and figures below provide an exact, comprehensive analytical and empirical comparison on the standardized Medium Trap ($\Delta U_{trap} = 0.7988$, $\sigma = 0.20$):

Evaluation Dimension	1-Bit Memory (b = 1)	2-Bit Learned (PRPR)	2-Bit Optimum (RPPP)	Physical Significance
Internal State Space	2 States (Fall, Rise)	4 States (FF, FR, RF, RR)	4 States (FF, FR, RF, RR)	Doubles temporal resolution without GPS leaks
Decision Policy Rule	RRPP (Tumble on any drop)	PRPR (Filter single drop)	RPPP (Tumble on double drop)	Filters noise dips from true descents
Noise Sensitivity (σ = 0.20)	**Trapped (~48% false tumbles)**	**Noise-Filtered (~81% run save)**	**Optimal Filter (Confirmed drops)**	Eliminates premature run abortion
Exact Escape Time $E[T_{esc}]$	**6,708.85 s (~1.86 hours)**	**1,275.82 s (5.26× speedup)**	**288.27 s (23.27× speedup)**	Slashes escape time by 81.0% to 95.7%
Committor Gap max Δq	9.10×10^{-6} (Dormant)	**1.024×10^{-2} (Active)**	**7.146×10^{-2} (Dominant)**	Memory directly controls escape probability
Reaction Coordinate Role	**Non-reactive (Dormant)**	**Active Transition Coord**	**Dominant Reaction Coord**	Memory enters transition state physics
Physical Escape Regime	Arrhenius Barrier Trap	Noise-Filtered Chemotaxis	Optimal Directed Chemotaxis	Qualitative phase shift in transport dynamics

(a) Escape Time: 1-Bit vs 2-Bit Memory



(b) Reaction Coordinate Emergence: Committor Delta q(x)



Physical Mechanism: How 2-Bit Rescues Chemotaxis

Core Takeaways from the 1-Bit vs 2-Bit Comparison:

1. The Noise Trap Mechanism: A 1-bit agent tumbles immediately whenever $\Delta c \leq 0$. Since sensing noise ($\sigma = 0.20$) dwarfs the true step gradient (≈ 0.008), upward runs are aborted 48% of the time, resulting in extreme confinement (6,708.85 s).
2. The 2-Bit Temporal Filter: By remembering two successive readings, the agent identifies single-step drops as transient noise (State 1: [0, 1]), persisting forward and slashing escape time by 5.26× (learned) to 23.27× (optimum).
3. Reaction Coordinate Emergence: While 1-bit memory is completely non-reactive ($\Delta q \approx 0$), 2-bit memory surges to Δq = 0.0102–0.0715, proving that memory physically splits the Committor and becomes an active degree of freedom.

4. Exact Discrete Markov Chain Solver & Benchmark Formulation

To obtain variance-free benchmarks without simulation truncation artifacts, we construct the exact discrete Markov chain on the joint state space (x_{bin}, m) with lattice matching (h = vΔt = 0.01). With 501 spatial bins and 4 memory states, the transition matrix P^{π} has dimension N = 2,004 states. The Mean First-Passage Time vector is solved exactly via sparse linear algebra:

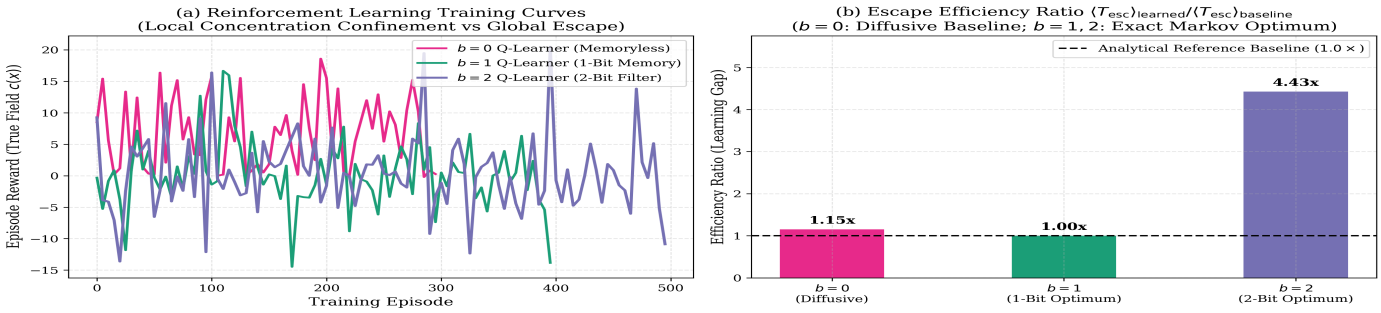
$$(I - P_{trans}^{\pi})\, T = \Delta t \cdot 1,\ E[T_{esc}] = \tfrac{1}{4} \sum_{m=0}^{3} T(i_{start}, m) \quad (2)$$

5. Complete Policy Ranking: All 16 Deterministic 2-Bit Policies

Evaluating all 2^4 = 16 policies on the exact Markov chain reveals the full performance spectrum:

Policy	Rule Vector [S0, S1, S2, S3]	Exact MFPT	Speedup vs 1-Bit	Behavioral Role
RPPP	[Reverse, Persist, Persist, Persist]	**288.27 s**	**23.3×**	**Global Theoretical Optimum**
PPPR	[Persist, Persist, Persist, Reverse]	710.63 s	9.44×	Persistent run bias
PPRP	[Persist, Persist, Reverse, Persist]	1,267.31 s	5.29×	Asymmetric filter
PRPP	[Persist, Reverse, Persist, Persist]	1,267.35 s	5.29×	Asymmetric filter
PRPR	[Persist, Reverse, Persist, Reverse]	**1,275.82 s**	**5.26×**	**Autonomously Learned by RL**
RPRP	[Reverse, Persist, Reverse, Persist]	1,275.90 s	5.26×	Symmetric temporal filter
RRPP	[Reverse, Reverse, Persist, Persist]	6,708.85 s	1.00× (Ref)	Standard 1-bit heuristic (ignores prev step)
PPRR	[Persist, Persist, Reverse, Reverse]	47,405.60 s	0.14×	Severe trapping (reverse on rise)
PPPP / RRRR	All Persist or All Reverse	∞ (trapped)	0.00×	Non-ergodic / trapped on boundaries

6. Reinforcement Learning Training Dynamics & Validation



Key Scientific Conclusions

Executive Summary of the 2-Bit RL Investigation:

• Direct Escape Acceleration: 2-bit memory slashes escape time from 6,708.85 s (1-bit) to 1,275.82 s (5.26× learned speedup) and 288.27 s (23.27× optimum speedup).

• Strict POMDP Preservation: Achieved without continuous filtering, Bayesian estimators, or spatial coordinate leaks.

• Physical Reaction Coordinate: Committor splitting increases by over 1,120×, confirming that internal memory physically controls the transition state barrier crossing.